Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2025

Course: B.Tech | *Time: 03 Hours*

Code: 103502 | *Subject: Wind and Solar Energy Systems* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.

(ii) There are NINE questions in this paper.

(iii) Attempt FIVE questions in all.

(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) The device used to convert wind energy into electrical energy is called
(i) Turbine
(ii) Transformer
(iii) Compressor
(iv) Inverter

(b) The theoretical maximum efficiency of a wind turbine is known as
(i) Carnot limit
(ii) Rankine limit
(iii) Betz limit
(iv) Joule limit

(c) The main material used in solar photovoltaic cells is
(i) Copper
(ii) Silicon
(iii) Aluminium
(iv) Iron

(d) A system combining wind and solar energy sources is called
(i) Thermal system
(ii) Mechanical system
(iii) Hybrid renewable system
(iv) Hydro system

(e) Which country is one of the largest producers of wind power in the world?
(i) Nepal
(ii) Sri Lanka
(iii) China
(iv) Pakistan

(f) The full form of DFIG is
(i) Double Field Induction Generator
(ii) Direct Flux Induction Generator
(iii) Doubly Fed Induction Generator
(iv) Dynamic Field Integrated Generator

(g) The converter used between generator and grid in variable-speed systems improves
(i) Noise level only
(ii) Tower height
(iii) Power quality
(iv) Blade length

(h) Pyranometer is used to measure
(i) Wind speed
(ii) Solar radiation
(iii) Temperature
(iv) Humidity

(i) Low Voltage Ride Through is associated with
(i) Battery charging
(ii) Water pumping
(iii) Wind and solar grid integration
(iv) Transformer cooling

(j) A grid code specifies
(i) Weather conditions
(ii) Solar panel color
(iii) Technical requirements for grid connection
(iv) Tower height only

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Q.2	(a)	Discuss the factors affecting wind power generation.	[7]
	(b)	Explain methods for estimation of solar energy availability.	[7]
Q.3	(a)	Explain the overview of grid code technical requirements for renewable energy systems.	[7]
	(b)	Discuss isolated operation of wind-solar hybrid systems with suitable block diagram.	[7]
Q.4	(a)	Explain Fresnel reflector technology with suitable diagram.	[7]
	(b)	Explain wind speed statistics and the importance of probability distributions in wind energy assessment.	[7]
Q.5	(a)	Explain the V-I characteristics of a PV cell with graph.	[7]
	(b)	Discuss worldwide experiences of renewable energy interconnection with power systems.	[7]
Q.6	(a)	Compare fixed-speed and variable-speed wind turbines.	[7]
	(b)	Explain voltage and frequency operating limits for grid-connected renewable systems.	[7]
Q.7	(a)	Discuss the advantages and disadvantages of parabolic trough systems.	[7]
	(b)	Describe the construction and operation of PV module and PV array.	[7]
Q.8	(a)	Explain different solar thermal energy technologies with neat diagrams.	[7]
	(b)	Explain Maximum Power Point Tracking and its importance.	[7]

Q.9 Write short note on ***any two*** of the following: **[7 x 2 = 14]**

(a) Explain Polycrystalline silicon
(b) Explain solar pond
(c) What is air density
(d) Define grid synchronization

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